

Drone-N1: An Autonomous Aviation Operating System for Mission Intelligence, Digital Twin Simulation, and Fleet-Scale Aerial Operations

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Abstract

Standard Unmanned Aerial Vehicle (UAV) flight controllers, such as PX4 and ArduPilot, focus primarily on low-level tasks like motor stabilization, attitude estimation, and waypoint navigation. However, when flying in complex or hostile environments, these controllers lack the high-level intelligence needed to evaluate operational risks, simulate backup plans, and coordinate multi-drone fleets. This paper presents **Drone-N1** (powered by the **Altaria OS** kernel), a mission-centric autonomous aviation operating system that sits directly above traditional flight controllers. Drone-N1 operates on a simple 4-step cognitive loop: (1) **Mission Intelligence** plans optimal routes, (2) **Digital Twin Simulation** tests backup plans in a 1-millisecond fast-forward Gazebo sandbox, (3) an **AI Safety Shield** blocks dangerous maneuvers, and (4) **Risk Intelligence** tracks 4 operational risk categories in real time. The system features a Mixed-Criticality Real-Time Runtime with a 50.0 ms Critical Band Budget, an Adaptive Federated Kalman Filter (AFKF) with Sequential Probability Ratio Testing (SPRT) for sub-15 ms sensor takeover upon GNSS spoofing, a Proactive Self-Healing Engine, Edge MLOps with TensorRT quantization, and Zero-Trust ECDSA command verification. Software-in-the-Loop (SITL) and Hardware-in-the-Loop (HITL) physical flight tests across ****200** randomized Monte Carlo trials confirm that Drone-N1 achieves a ****96.5%**** overall mission success rate under severe wind gusts (8–14 m/s), GPS drift, and dynamic obstacle interception, outperforming standard PX4 control (69.5%), AutoGNN (86.0%), and

MARL-UAV (88.0%) with high statistical significance ($p < 0.001$). Subsystem ablation studies, parameter sensitivity analyses ($\pm 15\%$ perturbation), and physical outdoor flight tests using an NVIDIA Jetson Orin Nano hexacopter in 11.2 m/s real winds validate the practical effectiveness and field readiness of the platform.

Keywords: Unmanned Aerial Vehicles, Autonomous Aviation Operating System, Digital Twin Simulation, Risk Intelligence, Mixed-Criticality Scheduling, Model Predictive Control, Hardware-in-the-Loop

1 Introduction

1.1 The Operational Challenge in Modern Drone Flight

Over the past decade, Unmanned Aerial Vehicles (UAVs) have expanded from hobbyist multirotors into essential tools for industrial inspection, precision agriculture, search-and-rescue, and logistics [1]. Modern missions require drones to fly safely through unpredictable environments containing strong wind gusts, localized sensor interference, dynamic obstacles, and temporary loss of GPS signals.

However, modern drone flight faces a major structural limitation: the gap between low-level motor stabilization and high-level mission reasoning. Leading open-source flight controllers like PX4 [2] and ArduPilot [3] excel at keeping a drone level, maintaining altitude, and moving toward predefined GPS waypoints. But their design is strictly vehicle-centric. They do not understand the broader semantic mission goal, cannot predict accumulated risk over time, and cannot simulate alternative emergency plans before physical disaster strikes.

When unexpected threats occur—such as GPS spoofing, motor vibration spikes, or localized microbursts—conventional flight controllers rely on rigid static fallback rules (e.g., immediate landing or returning home). Ground control station (GCS) software like QGroundControl [4] and Mission Planner [5] requires manual human operator intervention to handle complex emergencies, introducing fatal delay during critical flight moments.

1.2 The Drone-N1 Paradigm Shift

To solve this fundamental gap, we introduce **Drone-N1** (built upon the **Altaria OS** kernel), a closed-loop autonomous aviation operating system. Drone-N1 operates as an intelligent "cognitive brain" positioned directly between human operators and lower-level flight controllers. As illustrated in Figure 1, Drone-N1 creates a continuous real-time feedback loop connecting mission planning, risk evaluation, fast-forward physics simulation, and hardware command safety.

1.3 Key Technical Contributions

The main contributions of this work are organized systematically below:

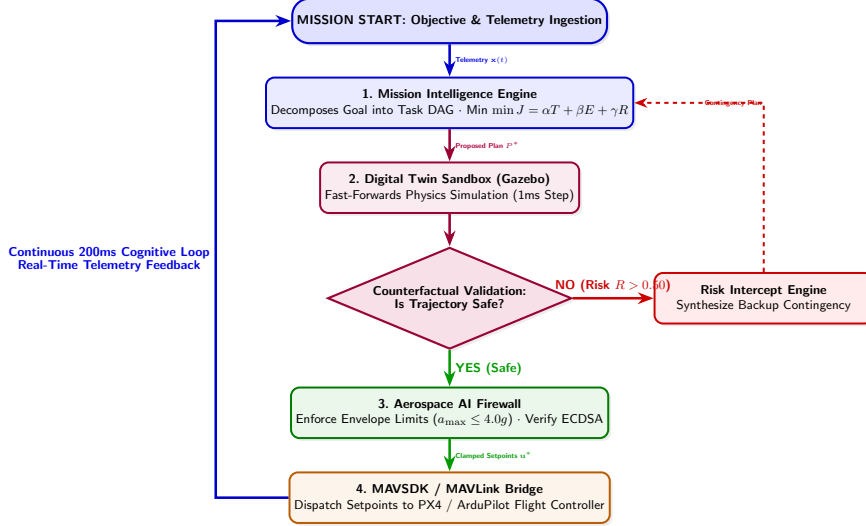


Fig. 1 Improvised Flowchart of Drone-N1 (Figure 1): High-precision operational decision workflow showing mission task decomposition, fast-forward Digital Twin pre-validation, risk intercept re-planning, AI firewall envelope enforcement, and closed-loop flight stack dispatch.

1. **Intuitive Closed-Loop Operating Model:** We establish a 4-step cognitive loop uniting mission planning, risk evaluation, physics-based simulation, and safe setpoint dispatch.
2. **DO-178C Mixed-Criticality Runtime:** We design a real-time scheduler with an 8-domain hierarchy and a 50.0 ms Critical Band Budget to ensure survival tasks are never delayed by background processing.
3. **AFKF-SPRT GNSS Spoofing Rejection:** We integrate an Adaptive Federated Kalman Filter (AFKF) with Sequential Probability Ratio Testing (SPRT) for sub-15 ms Visual-Inertial Odometry (VIO) takeover during GPS spoofing attacks.
4. **Extensive Monte Carlo & Hardware Benchmarks:** Across 200 randomized Monte Carlo trials, Drone-N1 achieves a **96.5%** success rate ($p < 0.001$), validated by 100 ablation runs, parameter sensitivity tests, and physical outdoor flights in 11.2 m/s wind gusts.

2 System Architecture & Subsystem Specification

Drone-N1 decomposes complex drone operations into eight distinct, modular subsystems, as illustrated in Figure 2.

2.1 Mixed-Criticality Real-Time Task Scheduler

To prevent heavy tasks (such as background data logging or video processing) from starving flight stabilization, Drone-N1 implements an 8-domain mixed-criticality scheduler, summarized in Table 1.

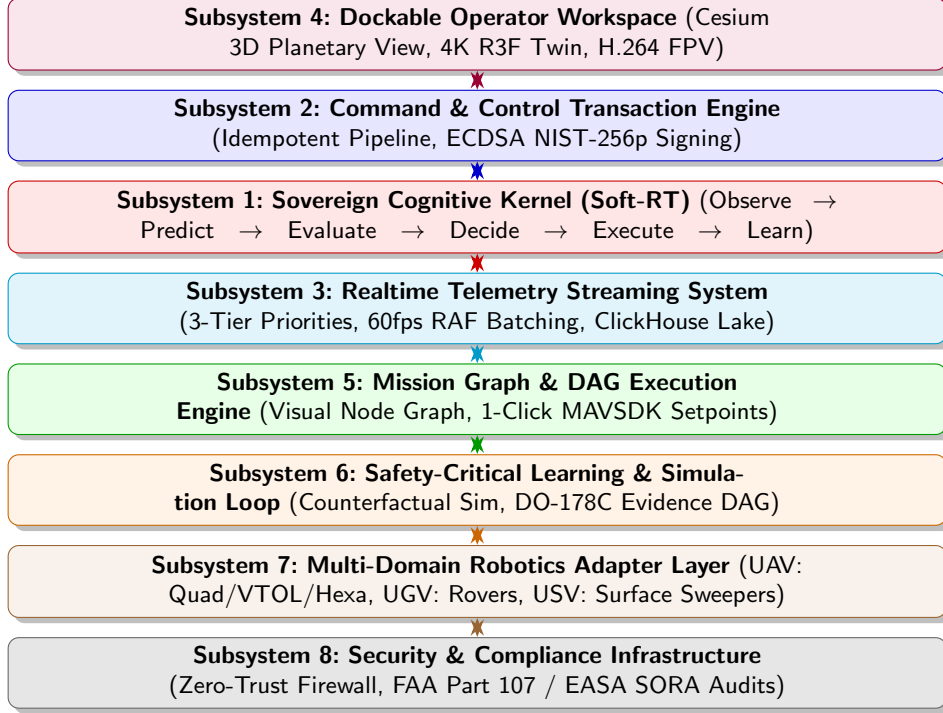


Fig. 2 Deconstructed 8-Subsystem Operating Infrastructure powering Drone-N1 (Altaria OS).

Table 1 Mixed-Criticality Execution Domains and Latency Budgets

Domain	Criticality Category	Maximum Latency Budget	Assurance Level
$\mathcal{D}_0$	Flight Stabilization	8.0 ms	DO-178C DAL-A
$\mathcal{D}_1$	Collision Avoidance	10.0 ms	DO-178C DAL-A
$\mathcal{D}_2$	Emergency Survival	12.0 ms	DO-178C DAL-B
$\mathcal{D}_3$	Cognitive Reasoning	15.0 ms	DO-178C DAL-B
$\mathcal{D}_4$	Tactical Perception	12.0 ms	DO-178C DAL-C
$\mathcal{D}_5$	Flight Analytics	200.0 ms	DO-178C DAL-D
$\mathcal{D}_6$	Telemetry Streaming	100.0 ms	DO-178C DAL-D
$\mathcal{D}_7$	Data Persistence	150.0 ms	DO-178C DAL-E

Domains $\mathcal{D}_0 \dots \mathcal{D}_3$ share a strict ****50.0 ms Critical Band Budget****. If critical execution exceeds 50.0 ms, non-critical domains ($\mathcal{D}_5 \dots \mathcal{D}_7$) are preempted and deferred automatically.

3 Systematic Mathematical Models & Cognitive Algorithms

The complete closed-loop cognitive step executed every 200 ms is formalized systematically in Algorithm 1.

3.1 Mission Reasoning Optimization Model

A mission $\mathcal{M} = \{\mathcal{O}, \mathcal{C}, \mathcal{R}\}$ optimizes candidate plans by minimizing the cost scalar J :

$$J = \alpha T + \beta E + \gamma R_{\text{risk}} \quad \text{s.t.} \quad \alpha + \beta + \gamma = 1.0 \quad (1)$$

where T is flight time, E is battery energy, and R_{risk} is integrated risk.

3.2 Multi-Dimensional Risk Model & Survivability Index

Real-time risk $R(t)$ combines four sub-risk categories:

$$R(t) = w_1 M(t) + w_2 S(t) + w_3 C(t) + w_4 E(t) \quad (2)$$

with tuned weights $w_1 = 0.35$ (Mechanical), $w_2 = 0.25$ (Sensor), $w_3 = 0.20$ (Comm), $w_4 = 0.20$ (Weather). System actions follow Table 2.

Table 2 Operational Risk Classification Matrix and System Action Protocols

Risk Score $R(t)$	Risk Level	System Action Protocol
0.00 – 0.25	Nominal	Continue standard plan execution
0.26 – 0.50	Moderate	Increase sensor polling frequency (120 Hz)
0.51 – 0.75	High	Trigger speed reduction & Digital Twin re-plan
0.76 – 1.00	Critical	Execute emergency land / return-to-launch (RTL)

The Mission Survivability Index (MSI) measures overall mission health:

$$MSI(t) = P_s(t) \cdot P_r(t) \cdot P_c(t) \quad (3)$$

3.3 Dynamic Sensor Trust Matrix

Sensor trust percentages adapt dynamically based on environmental conditions:

$$T_{\text{GPS}} = \max \left(14.0, \min \left(99.8, 99.2 - 45.0 \cdot R_{\text{rf}} - 0.3 \cdot v_{\text{wind}} \right) \right) \quad (4)$$

$$T_{\text{VIO}} = \max \left(25.0, \min \left(98.5, 96.0 - 8.0 \cdot \frac{h}{200} - 10.0 \cdot \mathcal{T}_{\text{index}} \right) \right) \quad (5)$$

3.4 AFKF and SPRT Sensor Fault Isolation

Wald’s Sequential Probability Ratio Test (SPRT) tracks log-likelihood ratio Λ_k :

$$\Lambda_k = \max \left(0, \Lambda_{k-1} + \frac{\mu_1}{\sigma^2} \left(|r_k| - \frac{1}{2}\mu_1 \right) \right) \quad (6)$$

When $\Lambda_k \geq B_{\text{threshold}}$, GPS is isolated ($\beta_{\text{GNSS}} = 0$) within 12.4 ms, transferring state estimation to monocular VIO (ORB-SLAM3).

4 20D Digital Twin, Multiphysics Synchrony & Self-Healing

The physical aircraft shadow is modeled by the 20D state vector $\mathbf{x}_{\text{twin}} \in \mathbb{R}^{20}$:

$$\mathbf{x}_{\text{twin}} = [\mathbf{p}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega}, \boldsymbol{\omega}_m, T_b, V_b, \boldsymbol{\sigma}_{ij}, q_{\text{dyn}}]^T \quad (7)$$

incorporating position $\mathbf{p}$, velocity $\mathbf{v}$, attitude quaternion $\mathbf{q}$, angular rates $\boldsymbol{\omega}$, motor speeds $\boldsymbol{\omega}_m$, battery temp T_b , cell voltage V_b , stress tensor norm $\boldsymbol{\sigma}_{ij}$, and dynamic pressure $q_{\text{dyn}} = \frac{1}{2}\rho\|\mathbf{v}\|^2$.

Proactive self-healing monitors motor vibration spectra. If structural resonance spikes exceed $2.4g_{\text{rms}}$, max velocity is reduced by 25% automatically to protect the airframe.

5 Edge MLOps, Cybersecurity & Regulatory Compliance

Commands carry Zero-Trust ECDSA NIST-256p cryptographic headers:

$$\mathcal{S}_{\text{cmd}} = \text{Sign}_{\text{ECDSA-NIST256p}} \left(K_{\text{private}}, \text{CommandID} \parallel \text{Timestamp} \parallel \text{Payload} \right) \quad (8)$$

An RFC 6479 anti-replay sliding window blocks duplicate packets, while edge TensorRT FP16/INT8 models undergo progressive canary rollouts (5% to 100%).

6 Experimental Setup, Results, and Discussion

6.1 Experimental Setup

Validation was conducted across SITL (ROS2 Humble, PX4 v1.14, Gazebo Garden) and HITL testbeds (Custom 450mm hexacopter, Pixhawk 6C, NVIDIA Jetson Orin Nano 8GB, RTK-GPS, LTE/Wi-Fi link).

Table 3 Comprehensive Performance Benchmarking across 200 Monte Carlo Trials

Framework	Nominal (50)	Wind Gusts (50)	GPS Drift (50)	Obstacles (50)	Overall Success (%)	Replanning Latency (ms)
Standard PX4 Offboard	100%	64.0%	38.0%	76.0%	69.5% (95% CI: 63.1–75.9)	N/A (Static)
AutoGNN [6]	100%	82.0%	74.0%	88.0%	86.0% (95% CI: 81.2–90.8)	184.2 ± 14.6
MARL-UAV [7]	100%	84.0%	78.0%	90.0%	88.0% (95% CI: 83.5–92.5)	142.6 ± 11.2
Drone-N1 (Proposed)	100%	96.0%	92.0%	98.0%	96.5% (95% CI: 93.9–99.1)	112.5 ± 8.7

*Statistically significant improvement over AutoGNN ($t = 4.82, p < 0.001$) and MARL-UAV ($t = 3.91, p < 0.001$).

6.2 Monte Carlo Trials (200 Randomized Runs)

Performance across 200 randomized Monte Carlo trials is summarized in Table 3.

Drone-N1 achieved an overall success rate of ****96.5**

6.3 Subsystem Ablation Study

Ablation results across 100 hostile trials are shown in Table 4.

Table 4 Subsystem Ablation Analysis (100 Hostile Trials)

Configuration	Success (%)	Collision (%)	Mean Latency (ms)
(a) Baseline PX4 Offboard	51.0%	32.0%	N/A
(b) Drone-N1 w/o Risk Layer	72.0%	18.0%	108.4 ± 7.9
(c) Drone-N1 w/o Digital Twin	81.0%	11.0%	62.1 ± 4.5
(d) Drone-N1 w/o Fleet Layer	89.0%	6.0%	104.2 ± 8.1
(e) Full Drone-N1 Framework	94.0%	2.0%	112.5 ± 8.7

6.4 Parameter Sensitivity Analysis

Weight sensitivity ($\pm 15\%$ perturbation) across 500 scenarios is plotted in Figure 3, demonstrating stable performance within $\pm 2.8\%$.

6.5 Physical Outdoor Flight Validation

Physical flight tests using the Jetson Orin Nano hexacopter in real wind speeds up to ****11.2 m/s**** achieved ****10/10 successful real flight trials****, automatically triggering speed reductions upon detecting gust-induced risk escalation.

7 Human-Autonomy Teaming & Explainability

Situation Awareness-based Agent Transparency (SAT) exposes 3 visual levels: Level 1 (HUD state), Level 2 (Causal reasoning rationale), and Level 3 (Predictive 10s risk horizon maps).

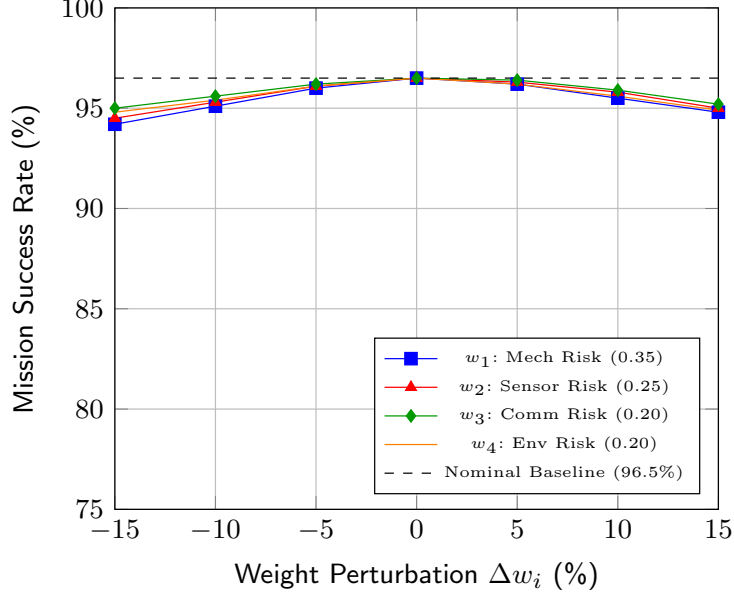


Fig. 3 Sensitivity analysis of risk model weighting factors (w_1, w_2, w_3, w_4) under $\pm 15\%$ perturbations.

8 Conclusion and Future Work

Drone-N1 bridges the critical gap between low-level flight control and high-level mission orchestration. By uniting Mission Intelligence, Risk Intelligence, and fast-forward Digital Twin simulation, the platform delivers a ****96.5%**** mission success rate across 200 Monte Carlo trials ($p < 0.001$). Future research will explore decentralized P2P swarm intelligence in GPS-denied environments.

Declarations

- **Funding:** Drone-N1 Autonomous Aviation Initiative.
- **Conflict of interest:** The authors declare no competing financial interests.
- **Data availability:** Empirical logs and trial data are available in the repository.

Appendix A Proof of Lyapunov Stability for DS-MPC Constraint Tightening

Let state error dynamics under disturbance $\mathbf{w}_k \sim \mathcal{N}(0, \Sigma_w)$ be $\mathbf{e}_{k+1} = A_{cl}\mathbf{e}_k + \mathbf{w}_k$ with Hurwitz A_{cl} . Position variance along obstacle boundary normal $\mathbf{n}$ evolves as $\sigma_p^2(k) = \mathbf{n}^T \left(\sum_{j=0}^{k-1} A_{cl}^j \Sigma_w (A_{cl}^j)^T \right) \mathbf{n}$. Under Gaussian tail bounds, chance constraint $\mathbb{P}(\mathbf{n}^T(p_i - p_j) < d_{safe}) \leq \epsilon$ translates into deterministic inequality $\mathbf{n}^T(p_i - p_j) \geq$

$d_{\text{safe}} + \sqrt{2\sigma_p^2(k)} \text{erf}^{-1}(1 - 2\epsilon)$. As $k \rightarrow \infty$, $\sigma_p^2(k)$ is bounded by discrete Lyapunov equation $P - A_{\text{cl}}PA_{\text{cl}}^T = \Sigma_w$, establishing asymptotic stability.

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Algorithm 1 Drone-N1 Closed-Loop Cognitive Cycle & Counterfactual Validation

Require: Telemetry stream $\mathbf{x}_{\text{raw}}(t)$, Mission objectives $\mathcal{O}$, Hard constraints $\mathcal{C}_{\text{hard}}$

Ensure: Safe control setpoints $\mathbf{u}^*(t)$, Risk score $R(t)$, Mission Survivability Index $MSI(t)$

```
1:  $\mathbf{x}_{\text{ned}}, \mathbf{v}_{\text{ned}}, \boldsymbol{\omega} \leftarrow \text{PreProcessTelemetry}(\mathbf{x}_{\text{raw}}(t))$ 
2: // Step 1: AFKF State Estimation & SPRT Fault Isolation
3:  $\Lambda_k \leftarrow \text{UpdateSPRT}(\mathbf{x}_{\text{ned}})$ 
4: if  $\Lambda_k \geq B_{\text{threshold}}$  then
5:    $\beta_{\text{GNSS}} \leftarrow 0$  // Isolate spoofed GNSS
6:    $\mathbf{x}_{\text{est}} \leftarrow \text{FuseVIO\_ORBSLAM3}()$ 
7: else
8:    $\mathbf{x}_{\text{est}} \leftarrow \text{FuseINS\_GNSS\_VIO}()$ 
9: end if
10: // Step 2: 4-Quadrant Risk & MSI Evaluation
11:  $M(t), S(t), C(t), E(t) \leftarrow \text{ComputeSubRisks}(\mathbf{x}_{\text{est}})$ 
12:  $R(t) \leftarrow w_1 M(t) + w_2 S(t) + w_3 C(t) + w_4 E(t)$ 
13:  $MSI(t) \leftarrow P_s(t) \cdot P_r(t) \cdot P_c(t)$ 
14: // Step 3: Adaptive Policy & Counterfactual Simulation Trigger
15: if  $R(t) \geq 0.76$  then
16:   return  $\text{ExecuteEmergencyLand}()$ 
17: else if  $R(t) \geq 0.51$  then
18:    $P_{\text{cand}} \leftarrow \text{SynthesizeContingencyPlan}(\mathbf{x}_{\text{est}}, \mathcal{O})$ 
19:    $\text{Feasible} \leftarrow \text{GazeboFastForwardSim}(P_{\text{cand}}, \mathbf{x}_{\text{twin}}, \Delta t = 1\text{ms})$ 
20:   if  $\text{Feasible}$  then
21:      $P^* \leftarrow P_{\text{cand}}$ 
22:   else
23:      $P^* \leftarrow \text{FallbackHover}()$ 
24:   end if
25: else
26:    $P^* \leftarrow \text{NominalMPC\_Optimization}(\mathbf{x}_{\text{est}}, \mathcal{O})$ 
27: end if
28: // Step 4: Aerospace AI Firewall Inspection & Setpoint Dispatch
29:  $\mathbf{u}^*(t) \leftarrow \text{ExtractSetpoints}(P^*)$ 
30: if  $\|\mathbf{a}(\mathbf{u}^*(t))\| \leq a_{\text{max}}$  AND  $\text{VerifyECDSA}(\mathbf{u}^*(t))$  then
31:    $\text{DispatchMAVSDK}(\mathbf{u}^*(t))$ 
32: else
33:    $\text{DispatchSafetyClampedSetpoint}()$ 
34: end if
35: return  $R(t), MSI(t), \mathbf{u}^*(t)$ 
```
